

This exam contains six problems; all problems have the same weight. The worst score will be dropped; only five solutions will be counted. Proofs or counterexamples are required for all problems. Time allowed: 2 hours 30 minutes.

Notation:  $\mathbb{R}$  is the real line;  $\mathbb{N}$  is the set of positive integers.

1. (a) Let  $I$  be an interval and let  $\{g_n\}_{n \in \mathbb{N}}$  be a sequence of real-valued functions on  $I$ . Give the definition of what it means for the sequence  $\{g_n\}_{n \in \mathbb{N}}$  to converge uniformly on  $I$  to a function  $g: I \rightarrow \mathbb{R}$ .  
(b) For each positive integer  $n$  let  $f_n: [0, 1] \rightarrow \mathbb{R}$  be defined by  $f_n(x) = (1 - x)^n$ . Prove that if  $0 < \tau < 1$ , then the sequence  $\{f_n\}_{n \in \mathbb{N}}$  converges uniformly on the interval  $[\tau, 1]$ .  
(c) Find  $f(x) = \lim_{n \rightarrow \infty} f_n(x)$  for  $x \in [0, 1]$ . Prove that the sequence  $\{f_n\}_{n \in \mathbb{N}}$  does not converge to  $f$  uniformly on the interval  $[0, 1]$ .
2. Let  $\{b_n\}_{n \in \mathbb{N}}$  be a sequence of real numbers. Suppose that there exists a real number  $\beta$  so that every subsequence of  $\{b_n\}_{n \in \mathbb{N}}$  has a further subsequence that converges to  $\beta$ . Prove that the entire original sequence  $\{b_n\}_{n \in \mathbb{N}}$  converges to  $\beta$ .
3. (a) Let  $S \subset \mathbb{R}$ . Give the definition of what it means for a function  $f: S \rightarrow \mathbb{R}$  to be uniformly continuous on  $S$ .  
(b) Prove or disprove the following statement: The function  $g(x) = x^2$  is uniformly continuous on  $\mathbb{R}$ .
4. The sumset of two sets  $A$  and  $B$  in  $\mathbb{R}^p$  is defined as:

$$A + B = \{a + b: a \in A, b \in B\}.$$

Prove that if  $A$  and  $B$  are non-empty compact sets in  $\mathbb{R}^p$ , then the set  $A + B$  is compact.

5. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a differentiable function such that  $\lim_{x \rightarrow \infty} f(x) = 0$ . Prove that there is a sequence  $\{c_n\}_{n \in \mathbb{N}}$  of real numbers such that  $\lim_{n \rightarrow \infty} c_n = \infty$  and  $\lim_{n \rightarrow \infty} f'(c_n) = 0$ .
6. Let  $I = [a, b]$  and let  $f$ ,  $g$ , and  $h$  be functions on  $I$  such that  $f(x) \leq g(x) \leq h(x)$  for all  $x \in I$ . Assume that  $f$  and  $h$  are Riemann integrable on  $I$  and that  $\int_a^b f(x) dx = \int_a^b h(x) dx = L$ .  
(a) Prove that  $g$  is Riemann integrable on  $I$ . (Hint: use the Riemann Criterion for Integrability.)  
(b) Show that

$$\int_a^b g(x) dx = L.$$